

Significant Santa Ana Winds/Wildfire Potential for Southern California Fall and Winter of 2025/2026

Last year, we witnessed historical wildfires fueled by dry fuels and damaging Santa Ana winds. Different atmospheric factors at the climatological level, both before and during the fall/winter last year led to an environment conducive for intense, fast-moving wildfires which sadly devastated the communities Pacific Palisades and Altadena. As we head into the next Santa Ana wind and wildfire season, what should we expect? In this article, we will take a look at three of the major influencing factors for significant wildfires- fuel moisture and quantity, winter rainfall, and the frequency and intensity of Santa Ana wind events.

Fuel Analysis

This summer, temperatures were near to slightly below average for much of the coastal areas of southern California. Toward the interior, temperatures were more near average to slightly above average. In terms of precipitation, the coastal and valley areas saw little precipitation from the summer monsoon, which is fairly normal even though the anomalies shown in Figure 1 show below-average percentages. For the mountains, the monsoon was near-average, with precipitation also coming in near-average.

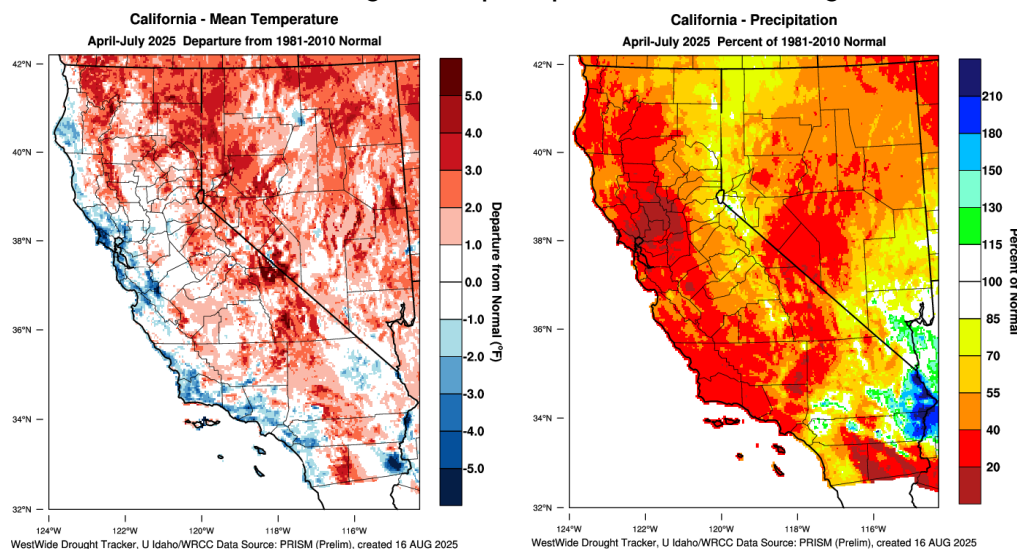


Figure 1: Temperature and Precipitation anomalies for the April-July 2025 period.

(Source: NOAA Regional Climate Centers, WestWide Drought Tracker)

These findings are supported by the fuel moisture graphs for the southern California coasts/valleys (Figure 2), eastern mountains (Figure 3), and southern mountains (Figure

4). Figure 2 shows near-average fuel moisture for the coast/valleys, and Figures 3 and 4 indicate below average fuel moisture for the mountains. While this is not directly correlated with the observed precipitation anomalies, these findings are acceptable given an anomalously warm summer in the mountains so far.

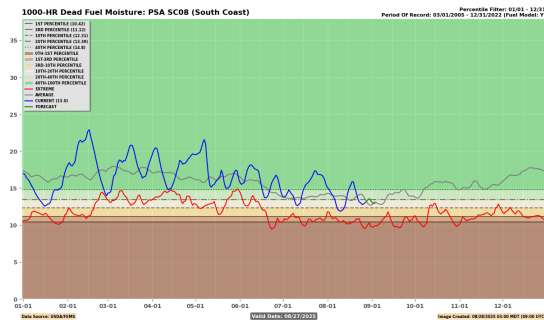


Figure 2: 1000-HR Fuel Moisture (bottom) for the Southern California coasts and valleys.

(Source: Southern California Geographic Coordination Center)

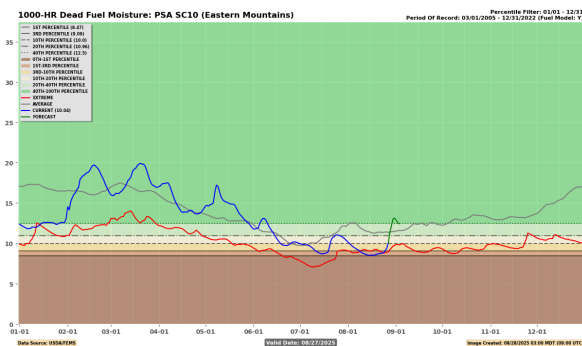


Figure 3: 1000-HR fuel moisture for the Los Angeles and San Bernardino County Mountains.

(Source: Southern California Geographic Coordination Center)

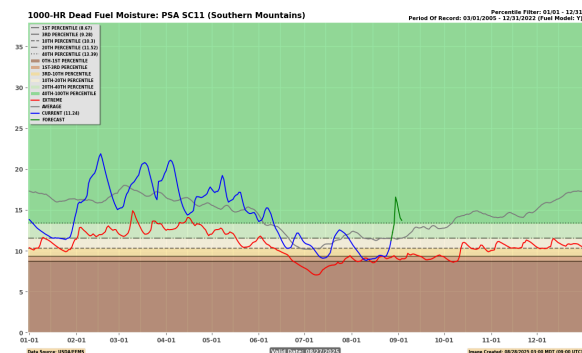


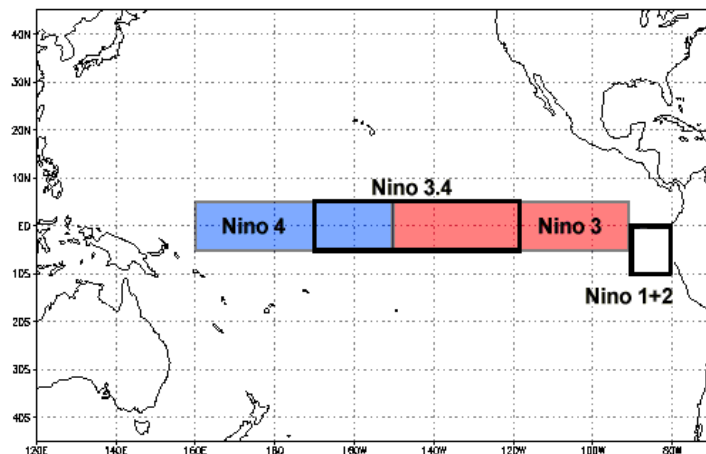
Figure 4: 1000-HR fuel moisture for the Riverside and San Diego County Mountains.

(Source: Southern California Geographic Coordination Center)

In terms of fuel quantity, a rather dry winter led to a limited green-up in the early Spring. However, dry fuels from previous years' winters (see Wildfire Forecast from Stroll November 2024) still remains. A consensus of this data leads to a near-average quantity of fuel present. Overall, both fuel quantity and fuel moisture is somewhat conducive for significant wildfires.

Global Climate Teleconnections

Now let's take a look at some Global Climate Teleconnections. Probably the most impactful of these Global Climate Teleconnections is the El Nino Southern Oscillation (or ENSO for short). El Nino (warm phase) occurs when the sea surface temperatures (SSTs) along the equatorial Pacific Ocean, specifically in the Nino 3.4 region, are 0.5 degrees Celsius or more above the seasonal average. Conversely, La Nina occurs when the SSTs along the equatorial Pacific Ocean in the Nino 3.4 Region are 0.5 degrees or more below normal. The different Nino regions are shown in Figure 5 below.



*Figure 5: The different Nino regions in the equatorial Pacific Ocean. ENSO is measured in the Nino 3.4 Region shown in the map above.
(Source: NOAA)*

Current ENSO status is ENSO-neutral with the latest SST anomalies being between -0.0 and -0.3 degrees Celsius, as shown in Figure 6.

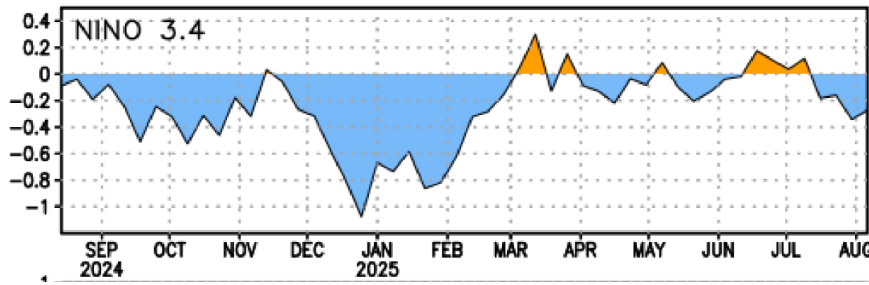


Figure 6: Current Nino 3.4 Region Sea Surface Temperature Anomalies. Latest observed values were between -0.0 and -0.3 degrees Celsius.
(Source: NOAA Climate Prediction Center)

The recent plume (Figure 7) of climate models indicate that SSTs will likely remain below average through this winter. A few of the more aggressive members indicate a weak La Nina developing early on this winter, then transitioning back to ENSO-neutral in the early Spring. However, both the dynamical and statistical averages keep ENSO conditions in the neutral phase, and majority of the members support this, so would expect below-average SSTs, but conditions remaining ENSO-neutral. In addition, the recent runs of the CFS climate model have been trending warmer closer to -0.0 anomalies, so this would add confidence in the ENSO-neutral prediction.

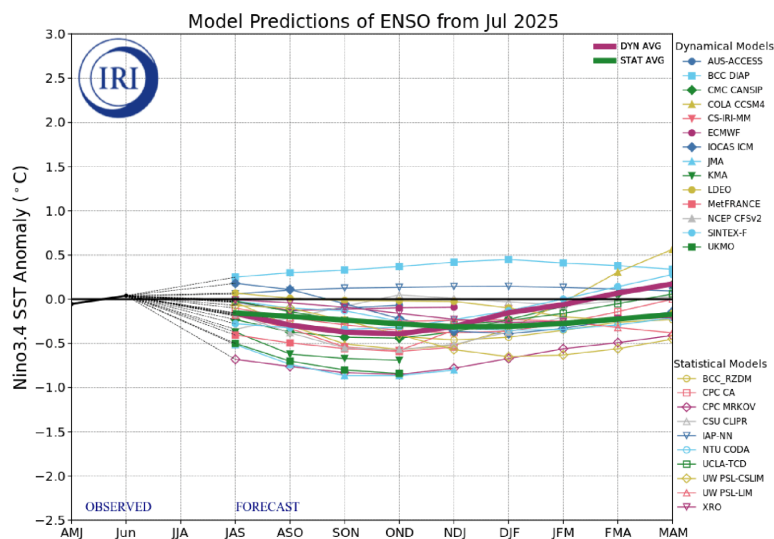


Figure 7: The recent IRI model predictions of ENSO into next year.

(Source: NOAA)

The ENSO-neutral forecast introduces uncertainty into the forecast, as neither the El Nino or La Nina weather pattern will dominate. Historically, southern California has seen both wet winters and dry winters during ENSO-neutral winters, making ENSO-neutral a difficult indicator to make a reliable forecast.

Another indicator that impacts intensity and frequency of Santa Ana winds is called the Pacific Decadal Oscillation (PDO). PDO analyzes the sea surface temperatures in the northern Pacific Ocean. During the positive phase of PDO, anomalously cool SSTs are present in the western and central portions of the northern Pacific basin and anomalously warm SSTs are present along the coast of Alaska and Canada. Conversely, negative PDO occurs when there are anomalously warm SSTs in the western and central portions of the northern Pacific basin and anomalously cool SSTs along the coast of Alaska and Canada. Typically, negative PDO, especially alongside a La Nina, results in more intrusions of Rossby waves, which are deep low pressure systems that occasionally break from the jet stream. When this occurs, it is known as a Rossby Wave-Breaking Event and is infamous for generating extreme weather. The premise for the damaging Santa Ana wind event January 7-9, 2025 that generated the Palisades and Eaton Fires was a Rossby Wave-Breaking event. Figure 8 shows upper air maps during the January 7-9, 2025 time period. Notice how the low pressure system denoted with the red “L”, is separated from the main jet stream denoted with the blue arrow. Because the system was so far removed from the jet stream, very little movement occurred and this allowed for an extended period of strong Santa Ana winds.

Find out more about the meteorology behind the January 7-9, 2025 Santa Ana wind event by going to the following hyperlink:

<https://www.maxweatherservice.com/presentations/santa-ana-winds-in-southern-california>

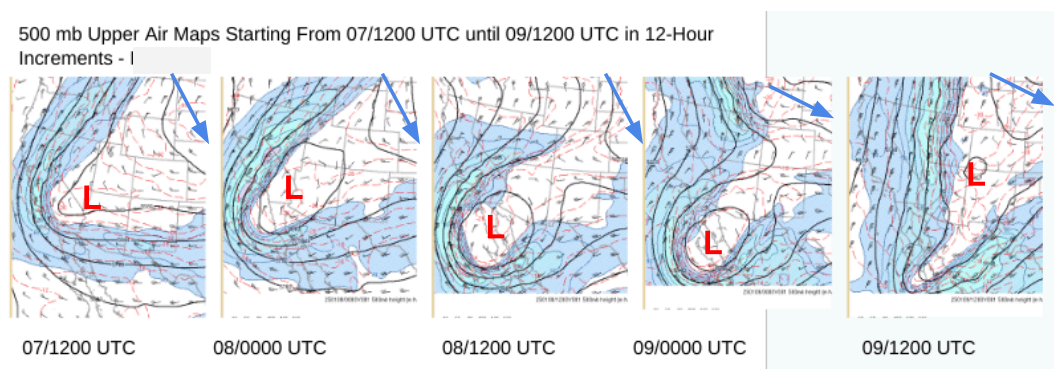


Figure 8: 500-mb winds and height contours during the January 7-9, 2025 Santa Ana wind event. Notice how the associated low pressure system denoted with the “L”, is separated from the main jet stream denoted with the blue arrow. This led a nearly stationary low pressure system over southern California, Nevada, and Arizona, bringing constant upper level support to fuel damaging Santa Ana winds.

For the past several years, we have been deeply entrenched in a negative PDO phase, as indicated in Figure 9, and it is likely to continue this way through the winter. Therefore, current and forecasted PDO is in favor of strong and more frequent Santa Ana wind events.

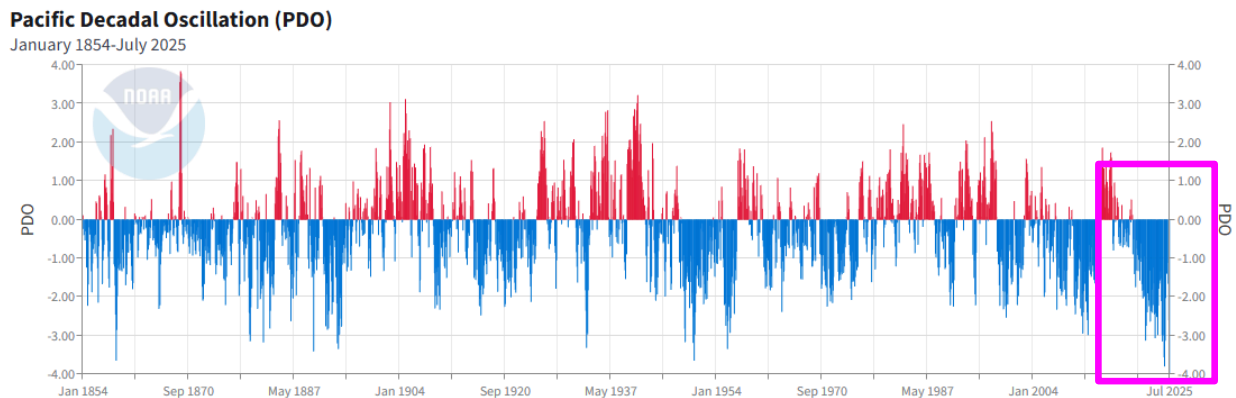


Figure 9: The Pacific Decadal Oscillation (PDO) history since 1854. The pink box shows that we have been in the negative phase of PDO over the last several years.

(Source: NOAA)

Finally, let's take a look at the Atlantic Multidecadal Oscillation (AMO). Similar to PDO, AMO analyzes SSTs in the northern Atlantic and is deeply tied to Atlantic Meridional Overturning Current or AMOC for short. Positive AMO occurs when SSTs in the North Atlantic are anomalously warm, and negative AMO occurs when SSTs in the Northern Atlantic are anomalously cool. Typically, positive AMO is more favorable for the development of Santa Ana winds. This is because positive AMO leads to subsidence (or sinking air) over the northeastern Pacific ocean and strengthens the North Pacific Subtropical High, a semi-permanent block of high pressure over the northeastern Pacific. When this system strengthens, it blocks incoming winter storms/rain from entering southern California, but allows the infamous "inside slider" path of low pressure systems that bring Santa Ana winds. Figure 10 shows the AMO index over the last 20 years. We have consistently been in a positive AMO phase over the last 20 years, with the AMO index increasing even further over the last 5 years or so.

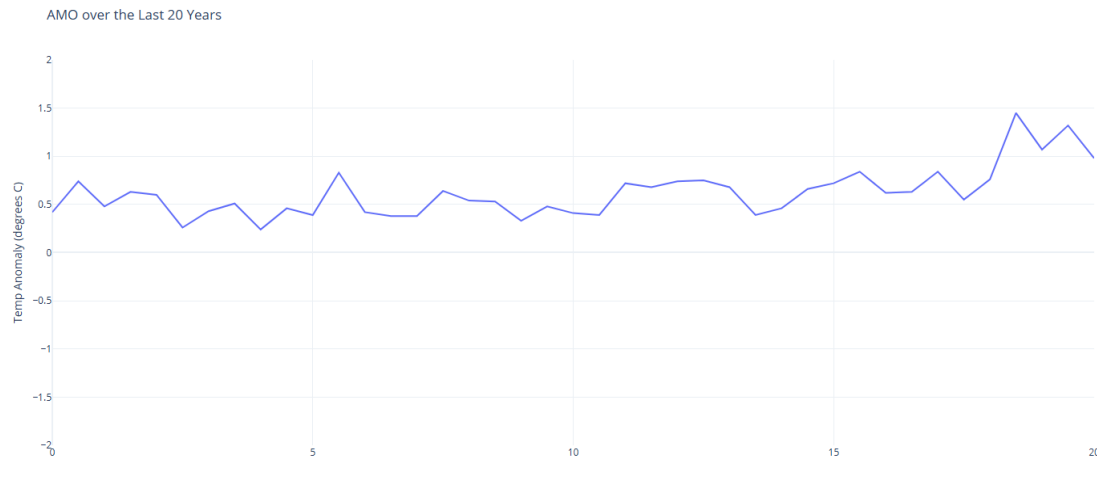


Figure 10: AMO index over the last 20 years.

Such anomalously high AMO values indicate that the environment is favorable for the strengthening of the North Pacific high, which synoptically inhibits rainfall and induces the development of Santa Ana winds. Note that this doesn't mean we won't get rainstorms, just the overall pattern is skewed toward favoring Santa Ana winds and drier days rather than rainstorms.

For more information on how AMO and PDO affect Santa Ana winds, be sure to check out *Santa Ana Winds: A Descriptive Climatology*, Rolinski et al. 2019 using the following hyperlink: https://journals.ametsoc.org/view/journals/wefo/34/2/waf-d-18-0160_1.xml?tab_body=fulltext-display and *The Influences of the Atlantic Multidecadal Oscillation on the Mean Strength of the North Pacific Subtropical High during Boreal Winter*, Lyu et al. 2017 using the following hyperlink: <https://journals.ametsoc.org/view/journals/clim/30/1/jcli-d-16-0525.1.xml>

Deterministic Climate Modeling/Official NOAA Outlooks

Alright, that's enough about Global Climate Teleconnections and theoreticals, let's take a look at some deterministic climate/long-range models!

First, let's take a look at NOAA's CFS climate model. Figure 11 shows the CFS model for precipitation anomaly during the Nov-Dec-Jan time period. As expected with the below-average SSTs, negative PDO, and positive AMO, conditions are favored for a below-average fall/winter for precipitation.

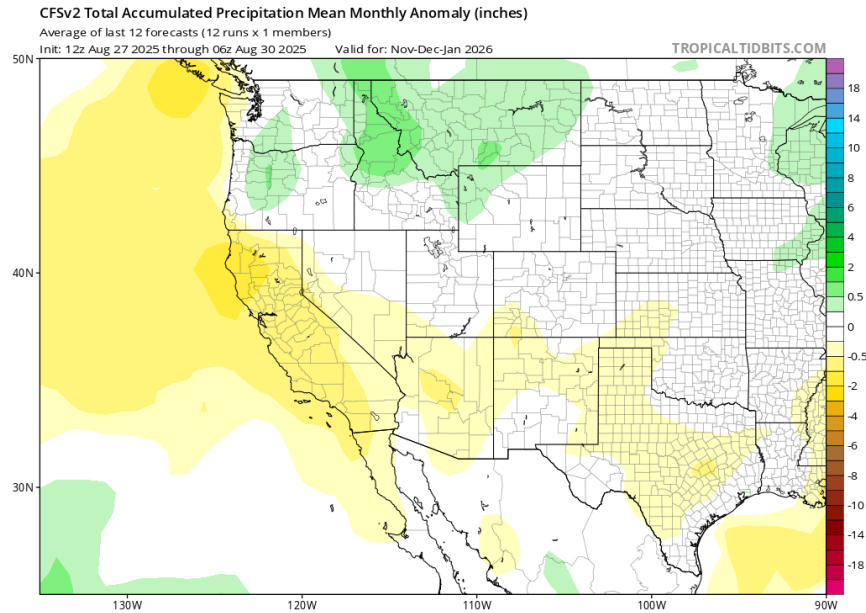


Figure 11: CFS Model forecast for Precipitation Anomaly Nov-Dec-Jan period

(Source: [TropicalTidbits.com](https://tropicaltidbits.com))

Next, let's see NOAA's official outlook for the Nov-Dec-Jan period. Figure 12 shows their three-month outlook for precipitation. Similar to the CFS model, the official outlook also outlines southern California in a below average contour. Also notice how the Pacific Northwest and Intermountain West have been outlined in an Above average contour which signals the presence of the "inside slider" pattern capable of Santa Ana winds.

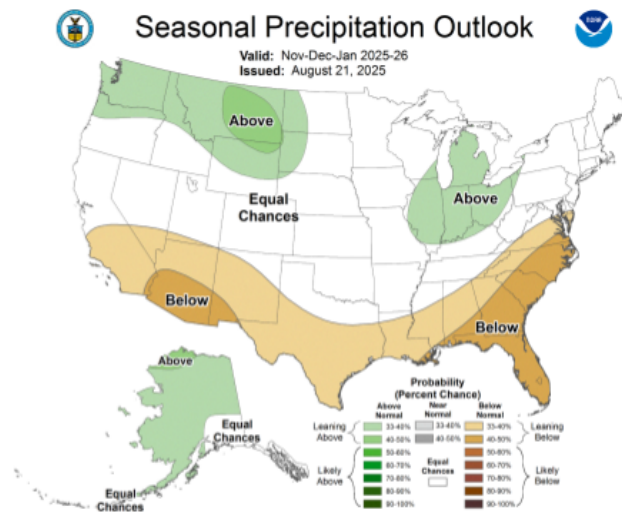


Figure 12: NOAA Climate Prediction Center's official three-month outlook for the Nov-Dec-Jan time period. Similar to the CFS model, southern California has been outlined in a below average contour.

(Source: NOAA)

Conclusion

Based on all available observed and forecasted data, including fuels, global climate teleconnections, and deterministic climate modeling, a near to slightly above average wildfire/Santa Ana wind season is likely, along with below-average precipitation. Both negative PDO and positive AMO contributes to a synoptic environment conducive for Santa Ana wind events, while deterministic climate modeling indicates drier than average conditions. Fuels are currently somewhat conducive for significant wildfires, although a dry next one or two months could allow for additional drying of fuels before wildfire season.

Figure 13 shows the National Interagency Fire Center's Significant Wildland Fire Potential Outlook which goes out to November. For November, above-normal significant wildfire activity is in the forecast. See their forecasts which are issued on the 1st of every month at the following hyperlink:

https://www.nifc.gov/nicc-files/predictive/outlooks/monthly_seasonal_outlook.pdf

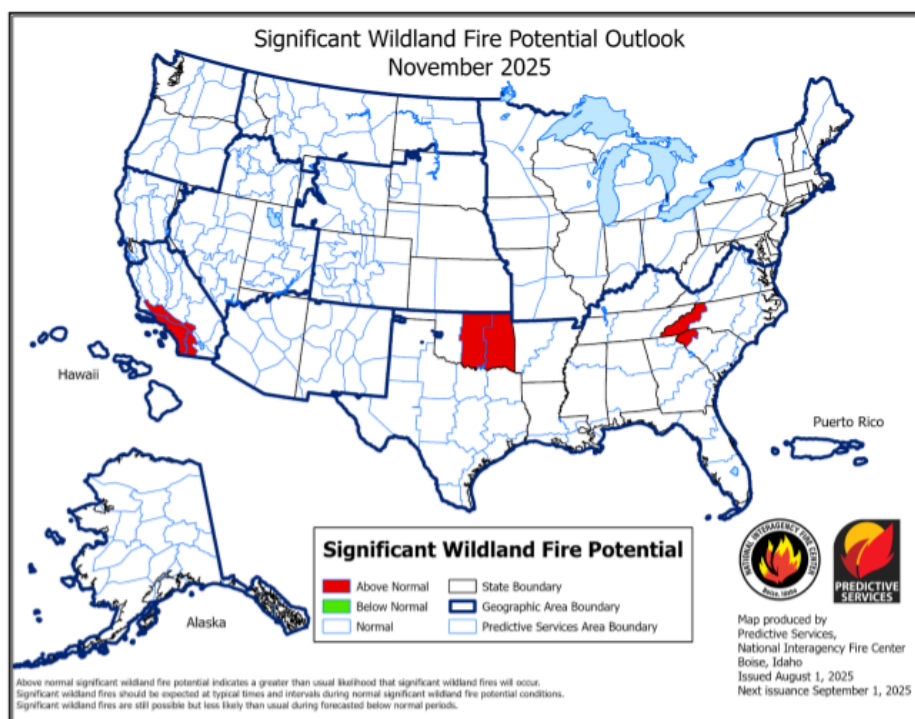


Figure 13: National Interagency Fire Center Significant Wildland Fire Potential Outlook valid for November 2025.

(Source: National Interagency Fire Center, Boise, ID)

References

NOAA Climate Prediction Center - www.cpc.ncep.noaa.gov

*Southern California Geographic Coordination Center
<https://gacc.nifc.gov/oscc/fuelsFireDanger.php>*

*Tropical Tidbits
www.tropicaltidbits.com*

*NOAA
www.noaa.gov*

*Santa Ana Winds: A Descriptive Climatology, Rolinski et al. 2019
https://journals.ametsoc.org/view/journals/wefo/34/2/waf-d-18-0160_1.xml?tab_body=fulltext-display*

*The Influences of the Atlantic Multidecadal Oscillation on the Mean Strength of the North Pacific Subtropical High during Boreal Winter, Lyu et al. 2017
<https://journals.ametsoc.org/view/journals/clim/30/1/jcli-d-16-0525.1.xml>*

*National Interagency Fire Center / Predictive Services
https://www.nifc.gov/nicc-files/predictive/outlooks/monthly_seasonal_outlook.pdf*